

Package ‘refweekreg’

June 18, 2026

Title Generate reference week regressors

Version 3.0

Description Generates reference week regressors to include in time series models.

These reference week regressors are indicator variables that have a 1 if the week a given US holiday occurs in is a reference week, 0 otherwise.

Routines also generate the day that starts the reference week (usually the week that contains the 12th of the month).

Another routine generates a 4 or 5 week regressor used to calendar adjust selected CES series.

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Encoding UTF-8

LazyData true

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Imports lubridate,
Rdpack,
stringr,
timeDate

Suggests seasonal

Depends R (>= 3.6)

RdMacros Rdpack

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calendar_mean_adj	<i>Calendar mean adjust regressor</i>
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Description

Remove the calendar month mean of a given regressor expressed as a time series object. Technical details and examples found in U. S. Census Bureau (2020).

Usage

```
calendar_mean_adj(this_reg = NULL)
```

Arguments

this_reg	Double precision time series array; a regressor from which the calendar month mean will be removed. This is a required entry.
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Details

Version 3.0, 2/9/2026

Value

Double precision time series array of the regressor with the calendar month mean removed.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

U. S. Census Bureau (2020). "Genhol - A utility that generates user-defined moving holiday regressors for X-13ARIMA-SEATS." <https://www2.census.gov/software/x-13arima-seats/win-genhol/download/genholdoc.pdf>. [Accessed 14-01-2026].

Examples

```
gf_years <- c(2001, 2006, 2017, 2022)
this_gf <-
  gen_rw_regressors(gf_years, 4, 2000, 2027, 'gf', remove_cal_means = FALSE)
this_gf_mean_adj <- calendar_mean_adj(this_gf)
```

difference_dates	<i>Date difference</i>
------------------	------------------------

Description

Generate the difference between two dates. Uses the `stringr` package (Wickham 2025).

Usage

```
difference_dates(first_date = NULL, second_date = NULL, this_frequency = 12)
```

Arguments

<code>first_date</code>	Integer array of length 2, a date where the first element is the year and the second element is the month or quarter. This is a required entry.
<code>second_date</code>	Integer array of length 2, a date to compare to <code>this_date</code> . This is a required entry.
<code>this_frequency</code>	Integer scalar, frequency of target time series. Entries limited to 1 (yearly), 2 (biannual), 3 (triannual), 4 (quarterly), 6 (Bimonthly), and 12. Default is 12 (monthly).

Details

Version 2.0 2/9/2026

Value

Integer scalar; difference between first date and second date.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Wickham H (2025). *stringr: Simple, Consistent Wrappers for Common String Operations*. R package version 1.6.0, <https://CRAN.R-project.org/package=stringr>.

Examples

```
diff_start <- difference_dates(start(shoes2007), c(1990,1))
```

```
gen_4to5_week_regressors
      generate 4/5 week regressors
```

Description

Generate 4 to 5 week effect regressors from paper by Cano et al. (1996). The result can be used with the `seasonal` package to include these effects in regARIMA models (Sax 2018; Sax and Eddelbuettel 2018).

Usage

```
gen_4to5_week_regressors(
  start_year = NULL,
  end_year = NULL,
  omit_march = TRUE,
  do_census_adj = FALSE,
  back_dates = NULL,
  remove_cal_means = FALSE
)
```

Arguments

<code>start_year</code>	First year of the sequence. This is a required argument.
<code>end_year</code>	Ending year of the sequence. This is a required argument.
<code>omit_march</code>	Logical scalar, exclude March from the set of regressors returned. Default is TRUE.
<code>do_census_adj</code>	Logical scalar, apply adjustment done by US Census Bureau for November and December. Default is FALSE.
<code>back_dates</code>	Array of date objects where the reference week needs to be set back a week by interviewers.
<code>remove_cal_means</code>	Logical scalar; if TRUE, calendar month means are removed from the final regression matrix. Technical details and examples found in U. S. Census Bureau (2020). Default setting is FALSE.

Details

Version 5.0 2/9/2026

Value

Matrix of time series arrays of 4/5 week regressors starting in January of `start_year` and ending in December of `end_year`. When `do_census_adj = TRUE`, the matrix returned has one column, otherwise the matrix is either 11 or 12 columns, with each column representing a different month.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Cano S, Getz P, Kropf J, Scott S, Stamas G (1996). “Adjusting for a Calendar Effect in Employment Time Series.” BLS Research Paper, U.S. Bureau of Labor Statistics. <https://www.bls.gov/osmr/research-papers/1996/pdf/st960190.pdf>.

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS*. R package version 1.7. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2020). “Genhol - A utility that generates user-defined moving holiday regressors for X-13ARIMA-SEATS.” <https://www2.census.gov/software/x-13arima-seats/win-genhol/download/genholdoc.pdf>. [Accessed 14-01-2026].

Examples

```
four2five_reg    <- gen_4to5_week_regressors(2005, 2024)
four2five_reg_cal <- gen_4to5_week_regressors(2005, 2024, remove_cal_means = TRUE)
replacement_dates <- c(as.Date('2013-11-10'), as.Date('2019-11-10'))
four2five_census_adj_reg <-
  gen_4to5_week_regressors(2005, 2024, do_census_adj = TRUE,
                           back_dates = replacement_dates)
```

gen_indirect_quarterly_holiday

Generate indirect quarterly holiday adjustments

Description

Generate indirect quarterly holiday factors and an indirect holiday adjusted series from monthly time series and monthly holiday factors.

Usage

```
gen_indirect_quarterly_holiday(this_a1 = NULL, this_hol = NULL, ratio = TRUE)
```

Arguments

this_a1	Real array; ts object of the original series. This is a required argument.
this_hol	Real array; ts object of the holiday factors. This is a required argument.
ratio	Logical scalar; if TRUE, holiday factors are assumed to be ratios; otherwise, the factors are assumed to be on the same scale as the original series. Default setting is TRUE.

Details

Version 2.1, 2/9/2026

Value

List object of two ts objects: `holadj`, which contains the indirect holiday adjusted quarterly series and `holfac`, the indirect holiday factors.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
n2033157_hol_q_list <- gen_indirect_quarterly_holiday(n2033157_a1, n2033157_hol)
```

```
gen_length_of_pay_period_reg
      generate length of pay period regressors
```

Description

Generate a version of the length of pay period regressor for monthly series used for some CES series. Uses the lubridate package (Grolemund and Wickham 2011).

Usage

```
gen_length_of_pay_period_reg(
  start_year = NULL,
  end_year = NULL,
  remove_cal_means = TRUE
)
```

Arguments

<code>start_year</code>	First year of the length of pay period regressors. This is a required argument.
<code>end_year</code>	Ending year of the length of pay period regressors. This is a required argument.
<code>remove_cal_means</code>	Logical scalar; if TRUE, calendar month means are removed from the final regression matrix. Technical details and examples found in U. S. Census Bureau (2020). Default setting is TRUE.

Details

Version 2.0 2/9/2026

Value

Matrix of a single time series array of a length of pay period regressors starting in January of `start_year` and ending in December of `end_year`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

U. S. Census Bureau (2020). “Genhol - A utility that generates user-defined moving holiday regressors for X-13ARIMA-SEATS.” <https://www2.census.gov/software/x-13arima-seats/win-genhol/download/genholdoc.pdf>. [Accessed 14-01-2026].

Examples

```
length_pay_reg_cal <-
  gen_length_of_pay_period_reg(2006, 2025, remove_cal_means = TRUE)
```

gen_reference_day	<i>Reference Day Generation</i>
-------------------	---------------------------------

Description

Generate the start of the reference week for a span of years, with an option to incorporate exceptions. For more details on reference weeks, see Bureau of Labor Statistics (2025) and Monsell (2021). Uses the lubridate package (Grolemund and Wickham 2011).

Usage

```
gen_reference_day(
  start_year = NULL,
  end_year = NULL,
  census_adj = TRUE,
  back_dates = NULL
)
```

Arguments

start_year	First year of the sequence of dates used to generate the reference day. This is a required argument.
end_year	Ending year of the sequence of dates used to generate the reference day. This is a required argument.
census_adj	Logical scalar, apply adjustment done by US Census Bureau for November and December. Default is TRUE.
back_dates	Array of date objects where the reference week needs to be set back a week by interviewers.

Details

Version 5.0, 2/9/2026

Value

Array of reference week dates starting in January of start_year and ending in December of end_year.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bureau of Labor Statistics (2025). “Concepts and Definitions (CPS) — bls.gov.” <https://www.bls.gov/cps/definitions.htm#refweek>. [Accessed 09-02-2026].

Grolemund G, Wickham H (2011). “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Monsell BC (2021). “Time Series Responses to the COVID-19 Pandemic at BLS for Monthly and Weekly Series.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*. <https://www.bls.gov/osmr/research-papers/2021/pdf/st210040.pdf>.

Examples

```
replacement_dates <- c(as.Date('2013-11-10'), as.Date('2019-11-10'))
rw2005 <- gen_reference_day(2005, 2024, back_dates = replacement_dates)
```

```
gen_reference_week_start_regressor
```

```
generate reference week start holiday regressors
```

Description

Generate reference week start holiday regressors, defined as a time series matrix. For more details on reference weeks, see Bureau of Labor Statistics (2025) and Monsell (2021). Uses the lubridate package (Grolemund and Wickham 2011).

Usage

```
gen_reference_week_start_regressor(
  this_reference_day = NULL,
  collapse_col = NULL,
  reg_means = NULL,
  contrast_reg = TRUE
)
```

Arguments

this_reference_day	Array of date objects of the start of the reference week for a given month. This is a required argument.
collapse_col	integer scalar; collapses the first collapse_col columns into a single column. Default setting is NULL, all columns are returned.
reg_means	numeric vector; vector of means to be removed from the regressors. Default is NULL, no mean removal done.
contrast_reg	Logical scalar; if TRUE, contrast regressors are generated. Default setting is TRUE.

Details

Version 5.1 2/9/2026

Value

Time series regression matrix object with reference week start regressors. The result can be used with the `seasonal` package to include these effects in regARIMA models (Sax 2018; Sax and Eddelbuettel 2018).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bureau of Labor Statistics (2025). “Concepts and Definitions (CPS) — bls.gov.” <https://www.bls.gov/cps/definitions.htm#refweek>. [Accessed 09-02-2026].

Grolemund G, Wickham H (2011). “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Monsell BC (2021). “Time Series Responses to the COVID-19 Pandemic at BLS for Monthly and Weekly Series.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*. <https://www.bls.gov/osmr/research-papers/2021/pdf/st210040.pdf>.

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS. R package version 1.7*. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
replacement_dates <- c(as.Date('2013-11-10'), as.Date('2019-11-10'))
rw2005 <- gen_reference_day(2005, 2024, back_dates = replacement_dates)
ref_week_start_reg <- gen_reference_week_start_regressor(rw2005)
```

gen_rw_holiday_matrix *Reference week regression matrix*

Description

Generate full regression matrix for reference week related holiday regressors, defined as a time series object. For more details on reference weeks, see Bureau of Labor Statistics (2025) and Monsell (2021). Uses the `lubridate` (Grolemund and Wickham 2011) and `timeDate` (Wuertz et al. 2025) packages.

Usage

```
gen_rw_holiday_matrix(
  this_reference_week = NULL,
  add_gf = TRUE,
  add_easter = TRUE,
  add_labor = TRUE,
  add_columbus = TRUE,
  add_vet = TRUE,
  remove_cal_means = TRUE
)
```

Arguments

this_reference_week	Array of date objects of the start of the reference week for a given month. This is a required argument.
add_gf	Logical scalar; if TRUE, a Good Friday holiday regressor will be included in the regression matrix. Default setting is TRUE.
add_easter	Logical scalar; if TRUE, an Easter holiday regressor will be included in the regression matrix. Default setting is TRUE.
add_labor	Logical scalar; if TRUE, a Labor Day holiday regressor will be included in the regression matrix. Default setting is TRUE.
add_columbus	Logical scalar; if TRUE, a Columbus Day holiday regressor will be included in the regression matrix. Default setting is TRUE.
add_vet	Logical scalar; if TRUE, a Veteran's Day holiday regressor will be included in the regression matrix. Default setting is TRUE.
remove_cal_means	Logical scalar; if TRUE, calendar month means are removed from the final regression matrix. Technical details and examples found in U. S. Census Bureau (2020). Default setting is TRUE.

Details

Version 3.0, 2/9/2026

Value

Array of reference week dates starting in January of start_year and ending in December of end_year The result can be used with the seasonal package to include these effects in regARIMA models (Sax 2018; Sax and Eddelbuettel 2018).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bureau of Labor Statistics (2025). "Concepts and Definitions (CPS) — bls.gov." <https://www.bls.gov/cps/definitions.htm#refweek>. [Accessed 09-02-2026].

Grolemund G, Wickham H (2011). "Dates and Times Made Easy with lubridate." *Journal of*

Statistical Software, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Monsell BC (2021). “Time Series Responses to the COVID-19 Pandemic at BLS for Monthly and Weekly Series.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*. <https://www.bls.gov/osmr/research-papers/2021/pdf/st210040.pdf>.

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS*. R package version 1.7. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2020). “Genhol - A utility that generates user-defined moving holiday regressors for X-13ARIMA-SEATS.” <https://www2.census.gov/software/x-13arima-seats/win-genhol/download/genholdoc.pdf>. [Accessed 14-01-2026].

Wuertz D, Setz T, Chalabi Y, Boshnakov GN (2025). *timeDate: Rmetrics - Chronological and Calendar Objects*. R package version 4051.111, <https://CRAN.R-project.org/package=timeDate>.

Examples

```
replacement_dates <- c(as.Date('2013-11-10'), as.Date('2019-11-10'))
rw2005 <- gen_reference_day(2005, 2024, back_dates = replacement_dates)
holiday_matrix_2005 <-
  gen_rw_holiday_matrix(rw2005, add_gf = FALSE, add_easter = FALSE,
                        remove_cal_means = FALSE)
```

gen_rw_regressors	<i>Generate reference week regressor</i>
-------------------	--

Description

Generate specific reference week related holiday regressors, defined as a time series object. The object returned is either a time series vector or a matrix of time series indicator variables. For more details on reference weeks, see Bureau of Labor Statistics (2025) and Monsell (2021).

Usage

```
gen_rw_regressors(
  holiday_years = NULL,
  holiday_month = NULL,
  start_year = NULL,
  end_year = NULL,
  col_label = "Reg",
  join_regressors = TRUE,
  remove_cal_means = TRUE
)
```

Arguments

holiday_years	Integer array; years where the holiday appears in the reference week (or for Columbus Day or Veteran's day, years where the holiday does not occur in the reference week). This is a required argument.
holiday_month	Integer scalar; the month in which this holiday occurs. For Good Friday and Easter, this would be April (4); for Labor Day, this would be September (9), etc. This is a required argument.
start_year	Integer scalar; first year of the generated holiday regressor. The regressor will begin on the first observation of this year. This is a required argument.
end_year	Integer scalar; final year of the generated holiday regressor. The regressor will end on the last observation of this year. This is a required argument.
col_label	Character scalar; a label used to generate column names for the individual AO regressors if join_regressors = FALSE.
join_regressors	Logical scalar; if TRUE, individual indicator regressors are combined into one grouped regressor. If FALSE, a matrix of the individual AO regressors will be returned. Default setting is TRUE.
remove_cal_means	Logical scalar; if TRUE, calendar month means are removed from the final regression matrix. Technical details and examples found in U. S. Census Bureau (2020). Default setting is TRUE.

Details

Version 4.1, 2/10/2026

Value

if join_regressors = TRUE, a time series object with the holiday regressor is returned; otherwise, a matrix of AO regressors for the individual holidays is returned. The result can be used with the seasonal package to include these effects in regARIMA models (Sax 2018; Sax and Eddebuettel 2018).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsel@gmail.com>

References

Bureau of Labor Statistics (2025). "Concepts and Definitions (CPS) — bls.gov." <https://www.bls.gov/cps/definitions.htm#refweek>. [Accessed 09-02-2026].

Monsell BC (2021). "Time Series Responses to the COVID-19 Pandemic at BLS for Monthly and Weekly Series." *Proceeding of the American Statistical Association, Business and Economic Statistics Section*. <https://www.bls.gov/osmr/research-papers/2021/pdf/st210040.pdf>.

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS. R package version 1.7*. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddebuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2020). “Genhol - A utility that generates user-defined moving holiday regressors for X-13ARIMA-SEATS.” <https://www2.census.gov/software/x-13arima-seats/win-genhol/download/genholdoc.pdf>. [Accessed 14-01-2026].

Examples

```
gf_years <- c(2001, 2006, 2017, 2022)
# returns a time series object with one grouped regressor
this_gf_grouped <-
  gen_rw_regressors(gf_years, 4, 2000, 2027)
# returns a time series matrix with four columns of indicator regressors
this_gf_individual <-
  gen_rw_regressors(gf_years, 4, 2000, 2027, 'gf', join_regressors = FALSE)
```

n2033157_a1	<i>At Work Series</i>
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Description

A time series object of an At Work Hour series.

Usage

```
n2033157_a1
```

Format

A time series object of an At Work Hours series published by the US Bureau of Labor Statistics from January of 2003 to May of 2020.

n2033157_ho1	<i>At Work Series Monthly Holiday Factors</i>
--------------	---

Description

A time series object of monthly holiday factors from an At Work Hour series

Usage

```
n2033157_ho1
```

Format

A time series object of monthly holiday factors from an At Work Hours series published by the US Bureau of Labor Statistics from January of 2003 to May of 2020.

save_user_reg	<i>save user regression matrix</i>
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Description

Save a user-defined regression array or matrix with time series attributes to an external ASCII file in X-13ARIMA-SEATS' datevalue format. For more information on the datevalue format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
save_user_reg(
  this_reg = NULL,
  this_reg_file = NULL,
  add_dates = TRUE,
  start_date = NULL,
  end_date = NULL
)
```

Arguments

this_reg	Double precision time series array or matrix to be saved. This is a required argument.
this_reg_file	Character string; name of file time series array or matrix to be saved to. This is a required argument.
add_dates	Logical scalar; save the file with dates on each record (datevalue format). Default is TRUE.
start_date	Integer vector of length 2, Start date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is start of series.
end_date	Integer vector of length 2, End date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is end of series.

Details

Version 5.0, 2/9/2026

Value

File with user-defined regressors will be produced.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Monsell BC, Lytras D, Findley D (2016). "Getting Started with X-13ARIMA-SEATS Input Files." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/gettingstartedx13-winx13.pdf>.

Examples

```
gf_years <- c(2001, 2006, 2017, 2022)
this_gf <-
  gen_rw_regressors(gf_years, 4, 2000, 2027, 'gf', remove_cal_means = FALSE)
## Not run: save_user_reg(this_gf, 'gf_2000_2027.txt')
```

shoes2007

Retail sales of shoes, ending in 2007

Description

A time series object of retail sales of shoes published by the US Census Bureau ending in December of 2007.

Usage

```
shoes2007
```

Format

A ts object of retail sales of shoes ending in December of 2007.

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